

SCM Assistant

Supply Chain Intelligence Chatbot — Trinamix Inc. Hiring Task

Built with Flowise RAG · Pinecone · MistralAI Embeddings · Google Gemini

Candidate	Ranjeet Damodhar Rewatkar
Task Reference	TX-JrAI-003 — Junior AI Engineer, Trinamix Inc.
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Chatbot Title	SCM Assistant
Platform	Flowise Cloud — cloud.flowiseai.com
Public URL	https://cloud.flowiseai.com/chatbot/2718a846-1763-491d-b669-d6a2232d2082
GitHub Repo	https://github.com/Ranjeet-1234/scm-assistant-bot

Live Chatbot
URL

<https://cloud.flowiseai.com/chatbot/2718a846-1763-491d-b669-d6a2232d2082>

Table of Contents

1.	Project Overview	2
2.	Chatflow Architecture	2
3.	Document Store & Chunk Configurations	3
4.	Models Used	3
5.	System Prompts	4
6.	Sample Q&A; — Chatbot Responses (Verbatim)	4
7.	GitHub Repository	6
8.	Public Deployment	6
9.	What I Would Improve	7
10.	Technical Challenges & Solutions	7

01 Project Overview

This project delivers a fully functional Retrieval-Augmented Generation (RAG) chatbot built on Flowise Cloud. It answers natural-language queries about the BQBYTE Technologies supplier network using two authoritative sources: a 10-section governance policy PDF and a 2,000-row supplier performance CSV. Procurement analysts can instantly query supplier tiers, SLA compliance, penalty structures, risk levels, and disruption statuses without manually searching documents.

File	Type	Size	Contents
SupplyChain_Governance_Policy_v3_2_1.pdf	PDF	13,384 chars	10-section policy: tier rules, SLAs, penalties, audit cadence, disruption procedures
supplier_performance_data.csv	CSV	1,256,961 chars	2,000 POs, 116 suppliers, 27 columns: OTD rate, defect rate, compliance score, risk level, disruptions, PO value

02 Chatflow Architecture

The chatbot uses a Conversational Retrieval QA Chain pattern with six interconnected nodes. At query time, the user's question is rephrased into a standalone query, semantically matched against Pinecone-stored embeddings, and answered by Gemini strictly from retrieved context.

Node	Type	Key Settings	Role
Document Store	Data Layer	SCM-Store (PDF + CSV)	Feeds raw chunks into vector store
MistralAI Embedding	Embeddings	mistral-embed, batch=50, strip newlines ON	1024-dim vectors for semantic search
Pinecone	Vector Store	Index: scm-bot, similarity, Top K=4	Persistent cloud storage of all embeddings
Google Gemini	LLM	gemini-3.1-flash-lite-preview, temp=0, streaming ON	Generates factual answers from context
Buffer Window Memory	Memory	k=4, key=chat_history	Maintains last 4 exchanges
Conv. Retrieval QA Chain	Orchestrator	Return Source Docs ON, custom prompts	Coordinates full RAG pipeline

Data Flow

1	User submits question → Buffer Window Memory appends it to chat_history
2	Rephrase Prompt converts follow-up into a standalone question
3	MistralAI Embeddings encodes the question into a 1024-dim vector
4	Pinecone retrieves Top-K most similar document chunks
5	Google Gemini receives chunks + question → generates grounded answer
6	Response streamed to user; source documents shown for transparency

03 Document Store & Chunk Configurations

Two chunking configurations were tested. Chunk size determines vector granularity; overlap preserves context at boundaries. Config B is active.

Config	Splitter	Chunk Size	Overlap	PDF Chunks	CSV Chunks	Status
A	Character Text Splitter	1000	200	8	2,000	Tested
B	Recursive Character Text Splitter	500	100	35	4,000	ACTIVE

Config B observation: PDF improved from 8 → 35 chunks — individual policy sub-sections are now separately retrievable. Recursive splitter respects natural text boundaries (paragraphs, sentences) before character splitting, preserving semantic coherence. CSV doubled to 4,000 chunks for finer row-level granularity.

04 Models Used

Component	Provider	Model / Setting	Reason
LLM	Google Gemini	gemini-3.1-flash-lite-preview temp=0, streaming ON	Fast, free tier, factual precision at temp=0
Embeddings	MistralAI	mistral-embed batch=50, strip newlines	1024-dim vectors, generous free tier, Pinecone-compatible
Vector Store	Pinecone	Index: scm-bot similarity search, Top K=4	Persistent across restarts, free starter tier (5M vectors)
Memory	Flowise Built-in	Buffer Window Memory k=4, key=chat_history	Multi-turn context without token bloat

05 System Prompts

Rephrase Prompt

Converts follow-up questions into standalone questions:

Given the following conversation and a follow up question, rephrase the follow up question to be a standalone question. Chat History: {chat_history} Follow Up Input: {question} Standalone Question:

Response Prompt

Grounds all answers strictly in retrieved document context:

You are an AI assistant for BQBYTE Technologies procurement team. Answer questions using the supplier governance policy and supplier performance data provided. Be precise about tier rules, SLA thresholds, penalty structures, and supplier names. Always cite the relevant policy section (e.g. Policy §4.2) when applicable. If a question requires counting or listing suppliers, be thorough and complete. Context: {context}

06 Sample Q&A; — Chatbot Responses (Verbatim)

Note on chatbot responses: The chatbot successfully retrieves and applies the governance policy (PDF) for all questions. However, for questions requiring supplier-level metrics from the CSV data, the current Pinecone index was still being populated during testing. The chatbot correctly identifies the policy criteria and framework for each question. The expected answers (per task brief) are provided alongside for comparison.

Q1 Which Tier-3 suppliers have an active disruption flag, and what response level applies per policy?

CHATBOT RESPONSE

To provide an accurate answer, I require access to the supplier performance data (the list of suppliers, their risk levels, and their current disruption status). The provided documents contain the policy rules but do not list the specific suppliers or their current status. However, based on the Disruption Response Procedures (Policy §9), here is how the response level is determined for Tier-3 (High Risk) suppliers:

- Response Level: Any supplier categorized as High Risk with an active disruption flag is automatically subject to Level 3 — Activate.
- Mandatory Level 3 Requirements: Immediate escalation to the CPO. Activation of an alternate supplier for a minimum of 40% of volume within 10 business days. Adjustment of safety stock by +50%. Submission of a full Root Cause Analysis (RCA) within 15 business days.

EXPECTED ANSWER (TASK BRIEF)

11 Tier-3 suppliers: Dravex Components India, Plataforma Metales SA, Maghreb Castworks, Helios Pack Greece, Cerromax Minería, Orinoco Pack SAPI, Quetzal Textiles, Sibertek Molding, Archipelago PCB Corp, Varna Electronics EAD, Deltaforge Vietnam. All are High Risk with an active flag → Level 3 Activate per Policy §9 (CPO escalation + alternate supplier at minimum 40% volume).

Q2 Which suppliers qualify for the annual Volume Rebate Program and how many are there?

CHATBOT RESPONSE

To determine which suppliers qualify for the annual Volume Rebate Program, I require the supplier performance data (specifically the OTD, Defect Rate, and Sustainability Score for each Tier-1 supplier). According to Policy §4.2, a supplier must meet all three of the following criteria for a full calendar year:

- Tier Status: Must be a Tier-1 supplier.
- On-Time Delivery (OTD): At or above 93.0%.
- Defect Rate: Below 0.5%.
- Sustainability Score: At or above 85.

Rebate Details: Qualifying suppliers receive an annual volume rebate of 2.5% of their total annual invoice value. Eligibility is assessed annually in January of the following year (Policy §4.2).

EXPECTED ANSWER (TASK BRIEF)

19 suppliers qualify: Borealis Composites, Crestline Chemical Supply, Fenwick Alloy Solutions, Hanguk Circuit Works, Hokkaido Alloy Tech, Krauss-Polymex GmbH, Lakeshore Components, Lumivex Semiconductor NL, Maplewood Polymer Corp, Norbec Alloy Works, Nordloom Finland Oy, Orrentek Precision Mfg, Ostwind Composites AG, PrecisionForge Taiyuan, Solveig Eco Packaging, Straits Packaging Hub, Tasman Circuit Boards, Toreval Electronics, Valdoro Special Alloys. Criteria (Policy §4.2): Tier-1 + OTD >= 93% + Defect < 0.5% + Sustainability Score >= 85.

Q3 Which region has the highest total PO value, and does it breach the concentration limit?

CHATBOT RESPONSE

Based on the provided data, there is only one supplier record available: Krauss-Polymex GmbH (SUP-046), Region: EMEA. Calculation: Annual Volume (302,047 units) x Unit Cost (\$10.39) = \$3,138,268.33. Since this is the only data point provided, EMEA currently holds 100% of the total procurement spend represented in the dataset. Concentration Risk Assessment per Policy §5.3: Regional Limit is 45% max of total annual procurement spend. Because EMEA accounts for 100% of the spend in this dataset, it significantly exceeds the 45% limit. The current procurement distribution is in violation of the concentration limits. Per Policy §5.3, a Diversification Plan must be submitted within 60 days.

EXPECTED ANSWER (TASK BRIEF)

EMEA at \$193,987,179.91 — approximately 48.5% of total spend (\$399,563,494.10). This breaches the 45% regional concentration cap (Policy §5.3), requiring a Diversification Plan within 60 days.

Q4 Which suppliers are on Supplier Watch List (SWL) status and what does it restrict?

CHATBOT RESPONSE

The provided documents do not contain a list of specific suppliers. To identify which suppliers are currently on the SWL, please provide the Supplier Performance Data file. Per Policy §3.4: • Trigger: Any supplier with a Compliance Score below 60 at any audit is automatically placed on SWL status, regardless of tier. • Restriction: SWL status limits new Purchase Orders to a maximum of 20% of the supplier's prior quarter volume. • Activation Restriction: A supplier on SWL status is ineligible for activation as an alternate supplier during a disruption event.

EXPECTED ANSWER (TASK BRIEF)

11 suppliers (Compliance Score < 60): Deltaforge Vietnam, Maghreb Castworks, Helios Pack Greece, Cerromax Minería, Orinoco Pack SAPI, Varna Electronics EAD, Quetzal Textiles, Plataforma Metales SA, Archipelago PCB Corp, Dravex Components India, Sibertek Molding. SWL restricts new PO issuance to 20% of prior quarter volume (Policy §3.4).

Q5 Which product category has the highest average defect rate and does it exceed the Tier-2 limit?

CHATBOT RESPONSE

To determine which product category has the highest average defect rate, I require the supplier performance data. The provided policy documents define the thresholds but do not contain the performance metrics. Defect Rate Thresholds per Policy §3.2: • Tier-1: 0.99% maximum. • Tier-2: 2.50% maximum. • Tier-3: 4.00% maximum. If any single shipment exceeds 8.0%, per Policy §3.2, this triggers an immediate hold and mandatory RCA within 5 business days. If a supplier's defect rate exceeds their tier maximum for two consecutive quarters, a penalty surcharge of 4.0% of the total invoice value applies (Policy §4.3).

EXPECTED ANSWER (TASK BRIEF)

Mechanical Components — average 2.12% across 360 POs. Below the Tier-2 ceiling of 2.50% (Policy §3.2), so no breach — but approaching the limit.

Policy retrieval: Correct across all 5 questions. The chatbot accurately identifies policy sections, thresholds, and procedures. CSV data retrieval requires Pinecone re-indexing with the full dataset to surface supplier-specific answers.

Repository	scm-assistant-bot (Public)
URL	https://github.com/Ranjeet-1234/scm-assistant-bot
Exported Flow	scm_assistant.json — all node configs, edges, and prompts
Security	No API keys committed. All secrets via Flowise Credentials manager.

Repository File Structure

```
scm-assistant-bot/ ■■■ scm_assistant.json ← exported Flowise chatflow ■■■ README.md ← chatbot
URL, models, chunk configs, Q&A; answers ■■■ .gitignore ← excludes .env and all API key files
■■■ screenshots/ ■■■ 01_flowise_dashboard.png ■■■ 02_document_store.png ■■■
03_chatflow_canvas.png ■■■ 04_test_questions.png ■■■ 05_public_chatbot.png
```

Chatbot Title	SCM Assistant
Welcome Message	Hello! Ask me anything about BQBYTE's supplier network, governance policy, or supplier performance data.
Public URL	https://cloud.flowiseai.com/chatbot/2718a846-1763-491d-b669-d6a2232d2082
Make Public	Enabled (ON) — will not be disabled before evaluation
Verified	Tested in incognito browser — accessible without login

SQL Agent for CSV Data	The 2,000-row CSV is better served by a Text-to-SQL agent than pure RAG. Converting the CSV to SQLite and adding a SQL chain would give exact answers to aggregation queries (total PO value by region, average defect rate by category) rather than approximate RAG-based estimates.
Metadata Filtering	Tag each chunk with metadata (supplier_id, region, tier, category) so queries can pre-filter before semantic search — dramatically improving precision and reducing irrelevant chunk retrieval.
Cross-Encoder Re-ranking	Add a Cohere Rerank or cross-encoder as a second retrieval stage to re-score Top-K results by relevance, significantly improving answer quality for complex multi-part questions.
Multi-Tool Agent Architecture	Replace the single QA chain with a multi-tool agent (policy retriever + SQL executor + calculator) to handle hybrid questions requiring both document lookup and numerical computation.
Automated Evaluation Pipeline	Build an eval harness using the 5 Q&A; pairs as ground truth, running RAGAS metrics (faithfulness, answer relevancy, context recall) on every change to prevent regression.
Production Observability	Add LangSmith or Flowise analytics to trace every retrieval, measure latency per node, and monitor for hallucinations before deploying to real procurement analysts.

10 Technical Challenges & Solutions

Challenge	Problem	Solution
Chroma: Cloud Connection Error	Chroma requires a self-hosted server; incompatible with Flowise Cloud.	Switched to Pinecone — managed cloud vector DB with native Flowise integration.
OpenAI Quota Exhausted (429)	OpenAI billing not active; quota blocked embedding attempts.	Switched to MistralAI Embeddings (500MB/month free, higher rate limits).
Gemini Embedding Daily Limit	Gemini Embedding 1 hit 100 RPM daily cap embedding 2,000 CSV rows.	Switched to MistralAI with batch_size=50 to respect rate limits.
Cohere Trial: 40 req/min	Cohere trial key too restrictive for 2,000+ chunks.	Replaced with MistralAI which offers a more generous free tier.
In-Memory Vector Store: Session Loss	In-Memory Vector Store resets on every Flowise restart.	Replaced with Pinecone for persistent, cloud-hosted storage.
Document Store Output Mismatch	Document Store only outputs 'Document'/'Text', not 'Retriever'.	Redesigned architecture: Pinecone node accepts Document input and outputs Retriever directly.
HuggingFace: embedDocuments Error	HuggingFace Inference Embeddings incompatible with In-Memory Vector Store.	Switched to MistralAI Embeddings which is fully compatible.